DISASTER RESPONSE ROBOTICS

GARUDA VEDANG

Autonomous Aerial Intelligence for the Golden Hour

PLATFORM ARCHITECTURE
GARUDA VEDANG
Two-tier mother-child aerial system

ENGINEERING FOCUS
Autonomous Avionics
Airframe, latch mechanics, edge CV

MISSION PROFILE
Disaster Scouting
Canopy penetration & survivor mapping

STATUS & CONTACT
TRL 2 Bench Validated
contact@garudavedang.aero

[TECHNICAL SCHEMATIC // CAD 1:10]

GARUDA VEDANG AIRFRAME
1.8m Fixed-Wing Carrier + Integrated Belly Scout

16:9 PROTOTYPE EMBED

The Disaster Information Void

Responders cannot deploy effectively without verified survivor coordinates

6–12 HRS

GROUND ACCESS DELAY

Roads & bridges cut off by floods

< 20 MIN

QUADCOPTER LIMIT

Battery exhausted before finding survivors

₹3.5L / HR

HELICOPTER SORTIE

Prohibitive cost & extreme rotor downwash

CURRENT STATUS QUO: BLIND SEARCH

- ✗ Speculative boat dispatches without verified coordinates
- ✗ Zero visibility beneath dense tree canopies and roofs
- ✗ Critical Golden Hour lost to navigation uncertainty
- ✗ Rescue assets delayed or deployed in incorrect sectors

[RESULT: > 80% SURVIVAL WINDOW COMPROMISED]

AERIAL INTELLIGENCE: TARGETED RESCUE

- ✓ 10 km river corridor mapped in a single aerial sortie
- ✓ Micro-scout drops directly under canopies to verify life
- ✓ Live GPS pins delivered straight to ground boat teams
- ✓ < ₹2,500 total operating cost per search sortie

[RESULT: < 15 MIN VERIFIED GPS DISPATCH]

Benchmarking Response Capabilities

Unresolved trade-offs between endurance, canopy inspection, and cost

PLATFORM	COVERAGE RADIUS	CANOPY INSPECTION	MISSION COST
Rescue Helicopters	100+ km radius	✕ Severe downwash hazard	✕ ₹3,00,000+ / hour
Consumer Drones	< 3 km line-of-sight	⚠ 20-min battery limit	⚠ ₹1.5L–₹3L fragile airframes
Ground Boat Teams	Delayed by flood debris	Direct visual, blocked paths	High crew risk & blind search
GARUDA VEDANG	**★ 10–15 km Wide Sweep**	**★ Dedicated Micro-Scout Drop**	**★ < ₹2,500 per mission**

HELICOPTERS
High reach • Prohibitive expense • Rotor downwash risk

SMALL DRONES
Agile camera • Severely constrained flight time • Small footprint

GARUDA VEDANG
Coordinated ecosystem • 10km scan + sub-canopy confirmation

Two-Tier Coordinated Reconnaissance

Wide-area fixed-wing scanning coupled with precision sub-canopy verification

TIER 01 // CARRIER

GARUDA MOTHERSHIP

1.8m WINGSPAN // 60-90 MIN ENDURANCE

[10 KM SWEEP | HIGH-LIFT DEPRON]

- Airframe: High-lift Depron composite structure
- Deployment: Internal belly bay holds micro-scout
- Telemetry: 868 MHz MAVLink command relay node

[TRL 2 BENCH VALIDATED // PASSIVE LATCH]

TIER 02 // SCOUT

VEDANG MICRO-QUAD

< 300g MASS // SUB-CANOPY PENETRATOR

[CANOPY DROP | OPTICAL FLOW HOVER]

- Separation: Passive magnetic gravity release
- Navigation: PMW3901 optical flow (GPS-denied)
- Downlink: 5.8 GHz live video streaming downlink

[500 CYCLES VERIFIED // SITL PROVEN]

TIER 03 // INTELLIGENCE

SCARLET-I EDGE AI

100% OFFLINE // YOLOv8 + THERMAL FUSION

[EDGE INFERENCE | REAL-TIME GPS MAP]

- Hardware: Rugged field ground station laptop
- Detection: Real-time person & flood classifier
- Dispatch: Instant Lat/Long rescue boat routing

[OFFLINE EDGE TESTED // ZERO CLOUD]

COORDINATED MISSION SEQUENCE

[1. 10KM FIXED-WING SWEEP] —> [2. MID-AIR SCOUT DROP] —> [3. SUB-CANOPY VERIFY] —> [4. GPS MAP TO BOATS]

Robust Engineering Architecture

Open-standard, fail-safe subsystems engineered for field reliability

SUBSYSTEM 01 // AIRFRAME

Aeromechanical Latch

500+

BENCH CYCLES

[0W CRUISE DRAW]

- Permanent magnet backup retention geometry
- Gravity-assisted drop separation without release servos
- Locked failsafe engagement during electrical power loss

SPECS: PASSIVE RELEASE // DUAL MAGNET BACKUP // FAILSAFE LOCKED

SUBSYSTEM 02 // GUIDANCE

GPS-Denied Autonomy

10/10

SITL TRIALS

[PMW3901 FLOW]

- Downward optical flow ground speed & position hold
- 6-axis IMU dead-reckoning stabilization in valleys
- ArduPilot open-standard GNC firmware integration

SPECS: PMW3901 SENSOR // 6-DOF IMU // ARDUPILOT OPEN STACK

SUBSYSTEM 03 // COMMS

Long-Range Telemetry

10+

KM RANGE

[868 MHz LINK]

- 868 MHz frequency penetrates rain and dense foliage
- 5.8 GHz low-latency live video streaming downlink
- Open MAVLink protocol command packet encryption

SPECS: 868 MHz MAVLink // 5.8 GHz VIDEO // 10+ KM LOS BUDGET

SUBSYSTEM 04 // DECISION

SCARLET-I Edge AI

100%

OFFLINE EDGE

[YOLOv8 FUSION]

- Real-time survivor & flood classifier on ground laptop
- Suppresses water sun glare & debris false alarms
- Zero cellular or cloud dependencies during outages

SPECS: OFFLINE YOLOv8 // THERMAL FUSION // RUGGED FIELD LAPTOP

Disciplined Bench Validation (TRL 2)

Empirical hardware testbench logging before outdoor flight trials

[TECHNICAL SCHEMATIC // CAD 1:10]

FULL-SCALE DEPRON TESTBED

1.8m Fixed-Wing Airframe • CG Balanced • Belly Latch Integrated

16:9 PROTOTYPE EMBED

500 / 500

LATCH CYCLES

100% release reliability

45–55 W

CRUISE POWER

Logged on thrust stand

10 / 10

SITL GPS-DENIED

Optical flow position hold

HARDWARE READINESS SUMMARY

- BENCH DEMONSTRATED: 500 latch releases, calibrated power draw, SITL code
- ENGINEERING TARGETS: 60–90 min outdoor endurance, full flight scout release
- NEXT MILESTONE: Transition from TRL 2 bench validation to TRL 4 flight test

[TESTBENCH VIDEO LOGS & TELEMETRY ARCHIVE]

Latching release videos, thrust bench logs, and ArduPilot SITL recordings available.

From Airborne Observation to Rescue Action

Real-time edge processing pipeline bridging aerial sensors to ground rescue

STAGE 01 // SENSING

Airborne Acquisition

OPTICAL + THERMAL + FLOW

[OPTICAL FLOW | 868 MHz TELEMETRY]

- GARUDA wide-area optical / thermal feed
- VEDANG sub-canopy tilt video link
- PMW3901 downward optical flow odometry
- 868 MHz MAVLink telemetry stream

[INTERFACE: MAVLink / 5.8GHz VIDEO]

STAGE 02 // COMPUTE

SCARLET-I AI Engine

OFFLINE YOLOv8 + FUSION

[PERSON & FLOOD CV | GLARE FILTER]

- Real-time person & flood classifier
- Dual optical & thermal cross-indexing
- Water glare & debris false alarm filter
- Instant coordinate georeferencing

[COMPUTE: Field Laptop CPU/GPU]

STAGE 03 // DISPATCH

Command Interface

GEO-TAGGED TACTICAL MAP

[SURVIVOR GPS | BOAT ROUTING]

- Offline interactive GIS situational map
- Verified Lat / Long GPS survivor pins
- Priority casualty triage classification
- Safe boat corridor navigation routing

[OUTPUT: QGIS / GeoJSON / Radio]

SAFETY ARCHITECTURE: DECISION SUPPORT ISOLATION

Flight control executes autonomously on real-time RTOS firmware. SCARLET-I runs strictly on the ground station to prevent computational latency from affecting flight safety.

Prioritized Market Segments

Deploying where faster situational awareness saves immediate lives

PHASE 01: BEACHHEAD

Disaster Response Agencies

NDRF, SDRF & District DDMAs

[700+ DISTRICT UNITS | EARLY ADOPTERS]

- Mission: Monsoon floods, landslides & cyclones
- Current Pain: Speculative blind boat deployment
- Solution: Direct survivor GPS delivered in < 15 min
- Adoption: Government disaster relief procurement

[INITIAL CONTRACT FOCUS // HIGH URGENCY]

PHASE 02: ADJACENT

Linear Utilities & Power

State Electricity Boards & Railways

[150K+ KM CORRIDORS | ANNUAL SCAN]

- Mission: Post-storm transmission line inspection
- Current Pain: Hazardous 2–3 day manual foot patrols
- Solution: 20 km powerline corridor scan in 1 flight
- Adoption: Enterprise infrastructure recurring utility contracts

[COMMERCIAL UTILITY MARKET // RECURRING]

PHASE 03: EXPANSION

Forestry & Wildlife Services

Forest Departments & Sanctuaries

[50+ NATIONAL PARKS | FIRE MONITORS]

- Mission: Wildfire perimeter tracking & anti-poaching
- Current Pain: Dense canopy hides smoldering hot-spots
- Solution: Micro-scout penetrates smoke to pinpoint fires
- Adoption: Central & State environmental surveillance tenders

[ECOSYSTEM EXPANSION // LONG-TERM]

ADOPTION PATHWAY

TRL 2 Bench Validation —> District Mock Drill Trials —> State Agency Pilots —> Fleet Scale Deployment

Measurable Operational Outcomes

Decisive reduction in search latency, operational expense, and responder risk

< 15 MIN

TIME TO FIRST PIN

Down from 6–12 hours of blind searching

> 90 %

DETECTION PRECISION

Optical and thermal cross-verification

< ₹2,500

MISSION COST

98% cheaper than manned helicopter sorties

OPERATIONAL DECISION ACCELERATION

- Eliminates speculative boat runs into hazardous flash flood rapids
- Directs emergency rescue aid while victims remain in the Golden Hour
- Replaces fragmented radio chatter with real-time map coordinates
- Optimizes battery and fuel consumption across all ground rescue assets

[MISSION GAIN: 40X FASTER SURVIVOR LOCALIZATION]

CREW SAFETY & ECONOMIC SCALABILITY

- Keeps rescue personnel safe from treacherous submerged obstacles
- Low unit cost enables every district headquarters to maintain units
- Zero fuel logistics burden compared to turbine-engine aircraft
- Scalable across district disaster authorities and municipal zones

[ECONOMIC MOAT: 98% SAVINGS VS MANNED HELICOPTER]

Sustainable Deployment Strategy

Hardware deployment paired with recurring software and support revenue

CAPITAL ACQUISITION

1. PLATFORM SALES

Carrier + Scout hardware airframes delivered to disaster agencies.

RECURRING SOFTWARE

2. SCARLET-I LICENSES

Annual subscription for CV updates, AI fine-tuning & tactical maps.

FIELD SUPPORT

3. SPARES & TRAINING

Operator training certifications, spare airframes & modular sensor pods.

MILESTONE METRIC	PHASE 1 (MONTH 1–6)	PHASE 2 (MONTH 7–12)	PHASE 3 (MONTH 13–24)
Hardware Units	2 Bench Testbeds	5 Field Pilot Systems	25+ Deployed Fleets
Software Accounts	Alpha Lab Instances	5 District Trial Accounts	25+ Annual Contracts
Target Milestone	**TRL 4 Integrated Flight**	**SDRF Exercise Simulation**	**State Disaster Procurement**

Engineering Risk Mitigation

Proactive technical mitigations across mechanical, avionic, and environmental domains

RISK 01: SEPARATION FAILURE

Mothership Release Jam

LOW PROBABILITY // HIGH IMPACT

[PASSIVE FAILSAFE BACKUP]

- Permanent magnet backup ensures separation during power loss
- Gravity-assisted drop design eliminates mechanical lockup
- 500+ bench cycles completed with 100% actuation success

[MITIGATION: 500 BENCH CYCLES // VERIFIED]

RISK 02: GPS NAVIGATION LOSS

Canopy Navigation Dropout

MEDIUM PROBABILITY // HIGH IMPACT

[OPTICAL FLOW + 6-DOF IMU]

- PMW3901 downward optical flow tracks ground velocity visually
- 6-DOF IMU dead-reckoning maintains position hold automatically
- Zero dependency on satellite lock while under dense forest cover

[MITIGATION: 10/10 SITL TRIALS // PROVEN]

RISK 03: MONSOON WIND TURBULENCE

Severe Gust Stability

HIGH PROBABILITY // MEDIUM IMPACT

[WING-LOADING ENVELOPE]

- Depron composite fixed-wing rated for 35 km/h continuous winds
- ArduPilot auto-abort trigger executes instant return-to-launch
- Low-altitude scout stays sheltered below canopy boundary layer

[MITIGATION: 35 KM/H ENVELOPE // BOUNDED]

BENCH VALIDATED

TRL 2 (CURRENT STATUS)

500 latch releases, calibrated thrust, SITL simulations

FIELD PROTOTYPE

TRL 4 (MONTH 6 TARGET)

Outdoor fixed-wing flight & mid-air scout deployment

RESCUE EXERCISE

TRL 6 (MONTH 12 TARGET)

Live flood response simulation with disaster authorities

Why GARUDA VEDANG Stands Apart

Resolving the fundamental trade-off between aerial endurance and inspection agility

01 // ARCHITECTURAL MOAT

DUAL-AIRFRAME SYNERGY

MOTHERSHIP + MICRO-SCOUT

[5X RECON REACH | CANOPY DROP]

- Fixed-wing covers 10+ km sectors while micro-quad penetrates canopy
- Eliminates the endurance vs. agility compromise of single drones
- Mothership remains at safe altitude while scout investigates

[BENCHMARK: 500% REACH VS SINGLE QUAD]

02 // ARCHITECTURAL MOAT

100% OFFLINE RESILIENCE

EDGE INFERENCE COMPUTING

[ZERO CLOUD NEED | REAL-TIME MAP]

- SCARLET-I operates with zero internet or cellular connectivity
- Full YOLOv8 inference runs on standard ground laptops in real time
- Zero data latency during communications infrastructure collapse

[BENCHMARK: ZERO NETWORK LATENCY]

03 // ARCHITECTURAL MOAT

ACCESSIBLE FIELD REPAIR

FRUGAL AEROSYSTEMS

[< ₹2,500 SORTIE | 15-MIN REPAIR]

- Depron XPS airframe uses off-the-shelf electronics and servos
- Damaged wings repaired in the field in under 15 minutes
- Sub-< ₹2,500 sortie economics vs ₹3,00,000+ for manned helicopters

[BENCHMARK: 98% SORTIE COST SAVINGS]

CORE ARCHITECTURAL DIFFERENTIATOR

A coordinated multi-tier ecosystem: Wide observation + Canopy inspection + Unified intelligence for disaster response.

Catalyzing the Transition: TRL 2 to TRL 4

Seeking ₹5–10 Lakh pre-seed support for integrated prototype flight trials

[+ PROJECT FOUNDER +]

Systems Architect

- Airframe design & Depron aero optimization
- Mechanical release latch & failsafe geometry
- 500+ bench actuation cycles verified

[+ AI & SOFTWARE LEAD +]

SCARLET-I Lead

- Edge YOLOv8 person & flood classifier
- Water glare & foliage false alarm filtering
- Offline GIS tactical map integration

[+ AVIONICS & GNC +]

Flight Firmware

- ArduPilot custom GNC & fail-safe RTOS
- PMW3901 optical flow odometry integration
- 868 MHz encrypted MAVLink telemetry

[+ VALIDATION LEAD +]

Test Engineer

- Thrust stand power draw calibration (45–55W)
- SITL 10/10 GPS-denied simulation trials
- Structural wind tunnel & drop testing

PRE-SEED CAPITAL REQUEST: ₹5 – ₹10 LAKH

- Airframe & Latch Machining (35%): Precision composite fabrication & servos
- Sensor & Optical Avionics (30%): Optical flow modules, telemetry & edge cameras
- Integrated Flight Tests (20%): Outdoor test range trials & wind endurance logging
- Provisional IP Documentation (15%): Patent filing on carrier-scout latching

INITIATE PARTNERSHIP

- EMAIL: contact@garudavedang.aero
- CODE: github.com/garuda-vedang-aerospace
- PHONE: +91 XXXXX XXXXX
- OBJECTIVE: TRL 4 Integrated Flight

[BENCH ARCHIVE]

BENCH LOGS

[Scan QR]